

OSCAR RAFAEL LAM HO

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EXECUTIVE SUMMARY

Data Engineer with hands-on experience building end-to-end pipelines using Airflow, Spark, Snowflake, and dbt, with expertise in batch and streaming architectures on AWS. Proven ability to deliver production-grade data solutions including star schema modeling, real-time streaming systems, and ML-integrated feature stores.

PROJECT EXPERIENCE

NYC 311 Service Equity Mart — [github link](#) | [live dashboard link](#)

Airflow · Snowflake · dbt · AWS S3 · Streamlit

- Built incremental Airflow DAG ingesting ~40M NYC 311 requests from Socrata API into S3, landing a raw → staging → mart pipeline in Snowflake with dbt (12 models, 38 tests).
- Modeled a star schema (fct_response_time × dim_tract × dim_complaint) joining 311 with ACS demographics; exposes response-time equity splits by borough and income quartile.
- Deployed Streamlit dashboard on Streamlit Community Cloud; full dbt docs site auto-published via GitHub Actions on merge to main.

Supply Chain Intelligence Engine — [github link](#) | [demo video link](#)

Python · PySpark · Dagster · Great Expectations · Plotly/Dash

- Orchestrated Dagster pipeline joining USDA NASS (production), USDA AMS (commodity prices), and FRED (macro) into a weekly parquet feature store with >60 engineered features (lags, rolling means, YoY deltas).
- Enforced data quality with Great Expectations at ingestion and post-join; schema drift caught before the feature store.
- Trained XGBoost price forecast (13-week horizon); exposed composite supply-risk score via an interactive Dash dashboard (gauge + radar + scenario panel), live at link.

MTA Subway Reliability Tracker — [github link](#) | [live API link](#)

Kafka (Redpanda) · Spark Structured Streaming · Postgres · S3 · FastAPI

- Designed a lambda architecture: GTFS-Realtime producer → Kafka, Structured Streaming consumer with 5-minute windowed aggregates, hot path to Postgres for a FastAPI endpoint, cold path to S3 Parquet.
- Computes on-time percentage per line and delay percentiles per station in near-real-time; full Docker Compose dev stack included.

TECHNICAL SKILLS

Languages:	Python, SQL (advanced: window functions, CTEs, MERGE, incremental patterns), Bash
Orchestration:	Airflow, Dagster
Processing:	Spark (DataFrames, Structured Streaming), Pandas, Great Expectations
Warehousing:	Snowflake (COPY INTO, Snowpipe, Time Travel), dbt
Cloud:	AWS (S3, Lambda, Athena, IAM, RDS)
Streaming:	Kafka / Redpanda, Spark Structured Streaming
Dev:	Docker, docker-compose, Git, GitHub Actions CI/CD, FastAPI, Streamlit, Plotly/Dash

WORK EXPERIENCE

Luckin Coffee, New York, NY

Dec 2025 – April 2026

Assistant Manager

- Trained and onboarded 2 new hires per week by developing structured weekly training plans and delegating tasks to build team productivity
- Analyzed 100+ drinks and modifications daily, tracking hourly cup counts and customer satisfaction metrics to optimize store performance
- Managed inventory counts for 50+ raw products on a daily and weekly basis to minimize waste and maintain supply continuity

EDUCATION

CUNY Brooklyn College

Feb 2025

B.S. Computer Science — GPA 3.83 / 4.0

Relevant coursework: Database Systems, Distributed Systems, Algorithms, Software Engineering

CERTIFICATIONS

- AWS Certified Data Engineer – Associate (Jun 2026)
- dbt Analytics Engineer Certification (Jul 2026)